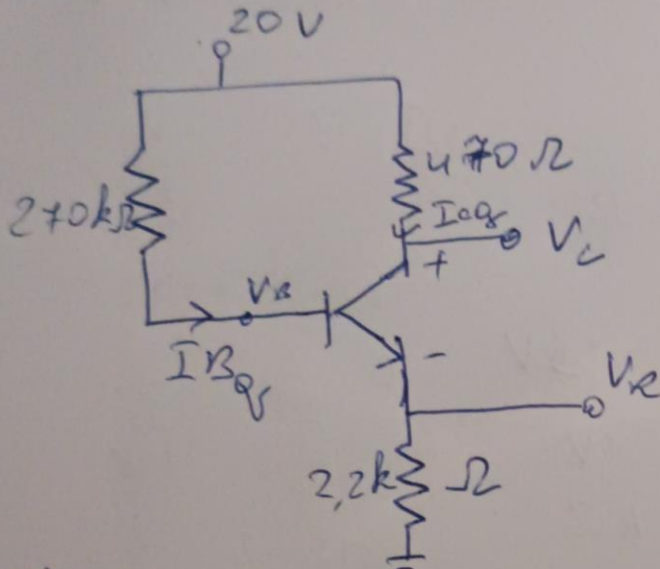




U 2210251 HW №8
D 245 ex 181



$$\begin{aligned} \text{a) } I_{BQ} &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 - 0,7}{270k + 126 \cdot 2,2k} = \\ &= \frac{19,3}{847,2k} = 35,27 \mu A \end{aligned}$$

$$\text{b) } I_{CQ} = \beta I_{BQ} = 125 \cdot 35,27 \mu = 4,41 \text{ mA}$$

$$\begin{aligned} \text{c) } V_{CEQ} &= V_{CC} - I_C(R_C + R_E) = 20 - (4,41 \text{ mA} \cdot \\ &\cdot (470\Omega + 2,2k)) = 8,23 \text{ V} \end{aligned}$$

$$\text{d) } V_C = V_{CC} - I_C R_C = 20 - 4,41 \cdot 470\Omega = 14,83 \text{ V}$$

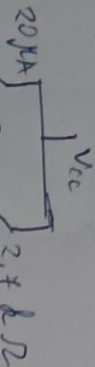




$$e) V_B = V_{CC} - I_B R_B = 20 - 35,2 \mu \cdot 210 k = 10,48 V$$

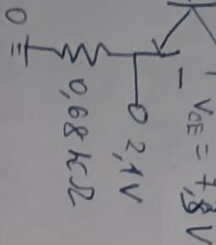
$$f) V_E = V_C - V_{CE} = 17,83 - 8,23 = 9,6 V$$

Page 24 of 24



$$a) I_C \approx I_E = \frac{V_E}{R_E} = \frac{9,6}{0,68 k} = 14,12 mA$$

$$b) V_{CE} = V_{RC} + V_{CE} + V_E = 15,5 V$$



$$c) V_{CE} = V_{RC} + V_{CE} + V_E = 15,5 V$$

$$d) V_C = V_{CC} - I_C R_C = 20 - 14,12 mA \cdot 210 k = 10,48 V$$

$$e) V_E = I_E R_E = 14,12 mA \cdot 0,68 k = 9,6 V$$

Page 24 of 24

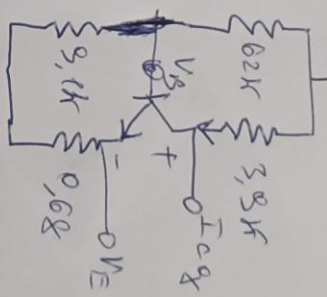
$$f) V_B = V_{CC} - I_B R_B = 20 - 35,2 \mu \cdot 210 k = 10,48 V$$

Exact approach

$$R_{TH} = R_1 \parallel R_2 = 4,84 k \Omega$$

$$I_{TH} = \frac{V_{CC} - V_{BE}}{R_{TH} + R_E} = 2,05 V$$

$$I_{BQ} = \frac{I_{TH} - V_{BE}}{R_{TH} + (\beta + 1) R_E} = 21,42 \mu A$$



$$b) I_{CQ} = \beta I_{BQ} = 1,71 mA$$

$$c) V_{CEQ} = V_{CC} - I_{CQ} (R_C + R_E) = 8,17 V$$

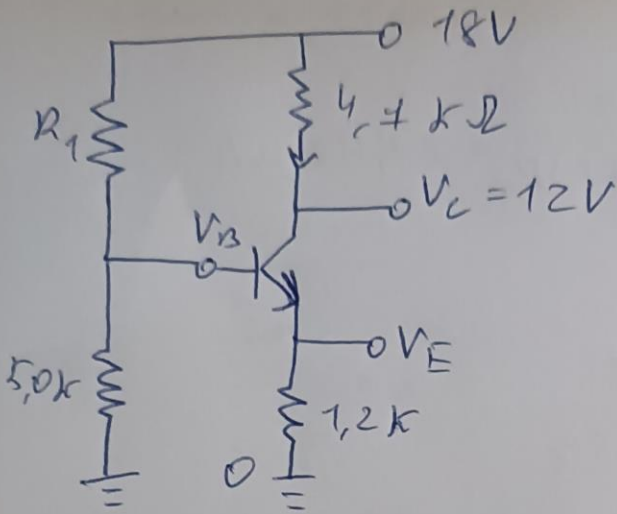
$$d) V_C = V_{CC} - I_C R_C = 9,33 V$$

$$e) V_E = I_E R_E = 1,16 V$$

$$f) V_B = V_{CC} - I_B R_B = 1,16 V + 0,4 V = 1,86 V$$



$$a) I_C = \frac{V_{CC} - V_C}{R_C} = \frac{18 - 12}{1,7 k} = 3,53 mA$$



$$b) V_E = I_E R_E \approx I_C R_E = 1,54V$$

$$c) V_B = V_{BE} + V_E = 0,4 + 1,54 = 2,24V$$

$$d) R_1 = \frac{V_{R1}}{I_{R1}}$$

$$V_{R1} = V_{CC} - V_B = 15,46V$$

$$I_{R1} \approx I_{R2} = \frac{V_B}{R_2} = 0,4mA$$

$$R_1 = \frac{15,46}{0,4mA} = 38,4k\Omega$$